

Yes, younger generations in Italy must adopt **supplementary pension savings and individual saving strategies** to maintain their retirement standard of living, but a full transition from PAYG to funded pensions faces **significant financial and political barriers**, with mixed evidence on the feasibility and welfare effects of a multi-pillar system.

1. Introduction

Italy's shift from a generous defined benefit (DB) pay-as-you-go (PAYG) pension system to a notional defined contribution (NDC) model has led to lower projected replacement rates for workers entering the labor market after 1995, raising concerns about retirement adequacy for younger generations (Franco & Tommasino, 2020; Mazzaferro, 2021; Borella & Moscarola, 2009). To compensate, policy reforms have promoted supplementary pension pillars—occupational and individual funded schemes—alongside voluntary private savings (Hinrichs, 2021; Capuano, 2015; Cesari et al., 2007; Pizzuti, 2022). However, participation in these supplementary schemes remains limited, and many young Italians lack awareness or motivation to save adequately for retirement (Bongini & Cucinelli, 2019; Rinaldi, 2011). The literature also debates the feasibility of transitioning from PAYG to funded systems: while mixed or multi-pillar approaches can diversify risks and potentially improve sustainability, they introduce transition costs and the "double burden" problem for the transition generation, with significant distributive and political challenges (Holzmann, 2017; Castellino & Fornero, 1999; Forni & Giordano, 2005; Ebbinghaus, 2015). International comparisons suggest that while diversification is desirable, successful transitions require careful design to balance adequacy, sustainability, and intergenerational equity (Boado-Penas et al., 2019; Poškutė et al., 2022).

Can supplementary pension savings and individual saving strategies help Italian workers maintain their standard of living at retirement under the NDC PAYG system?

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

FIGURE 1 Consensus meter: Can supplementary pension savings help maintain retirement standards under Italy's NDC PAYG system?

2. Methods

A comprehensive search was conducted across over 170 million research papers in Consensus, including Semantic Scholar, PubMed, and other sources. A total of 2,641 papers were identified; after relevance screening and deduplication, 50 papers were included in this review.

Search Strategy

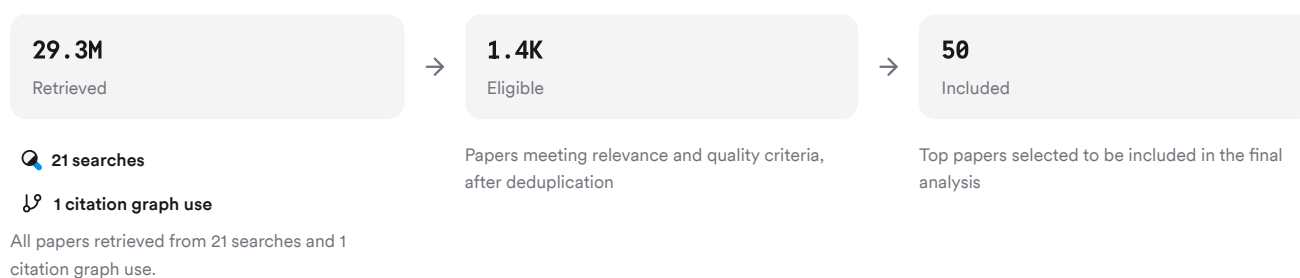


FIGURE 2 Flowchart of paper selection process for this review.

Six unique search strategies targeted foundational context, individual saving behaviors, alternative terminology (multi-pillar/occupational pensions), transition feasibility critiques, international comparisons, and adjacent demographic/economic topics.

3. Results

3.1 Individual Saving Strategies & Supplementary Pension Behaviors

Younger Italian workers are increasingly encouraged to participate in supplementary pension funds (second/third pillar), such as occupational or individual defined contribution plans (Capuano, 2015; Cesari et al., 2007; Pizzuti, 2022). Simulations show that combining public NDC pensions with supplementary funds can significantly increase replacement rates compared to relying solely on public pensions (Marchionne, 2021; Cesari et al., 2007). However, actual participation remains low due to limited financial literacy, lack of awareness about future pension shortfalls, and overestimation of public pension adequacy among young people (Bongini & Cucinelli, 2019; Rinaldi, 2011). Financial education and targeted policy interventions are recommended to boost voluntary savings (Bongini & Cucinelli, 2019).

3.2 Feasibility of Transitioning Toward Funded or Mixed Systems

The literature highlights both potential benefits and substantial obstacles in moving from a pure PAYG system toward a partially or fully funded model. While a mixed system could theoretically improve sustainability by diversifying funding sources and reducing demographic risk exposure (Castellino & Fornero, 1999; Boado-Penas et al., 2019), transition costs—including the need for current workers to finance both existing retirees (PAYG) and their own future benefits (funded)—are significant (Holzmann, 2017; Forni & Giordano, 2005; Ebbinghaus, 2015). Political resistance is strong due to the double burden on the transition generation and concerns about fairness (Ebbinghaus, 2015).

3.3 Transition Costs & Double Burden Problem

Transitioning from PAYG to funded systems makes implicit debt explicit: current workers must pay for both current retirees' benefits (via taxes/contributions) and accumulate assets for their own retirement (Holzmann, 2017; Andersen et al., 2021; Forni & Giordano, 2005). This "double burden" is often cited as a major barrier; some models suggest it can be mitigated through gradual transitions or by leveraging economic growth effects (Holzmann, 2017; Andersen et al., 2021), but empirical evidence shows that most countries avoid abrupt shifts due to these costs (Lin et al., 2021; Ebbinghaus, 2015).

3.4 Welfare Effects of Mixed Pillar Structures

Mixed systems—combining PAYG/NDC with mandatory or voluntary funded components—can offer better risk diversification but may result in lower overall replacement rates unless supplemented by additional savings or higher contributions (Boado-Penas et al., 2019; Devolder et al., 2020). Some cohorts may benefit more than others depending on career profiles and timing of reforms; distributional impacts must be carefully managed to avoid increasing old-age poverty risk among those with interrupted careers or low lifetime earnings (Hinrichs, 2021; Mazzaferro, 2021).

Results Timeline

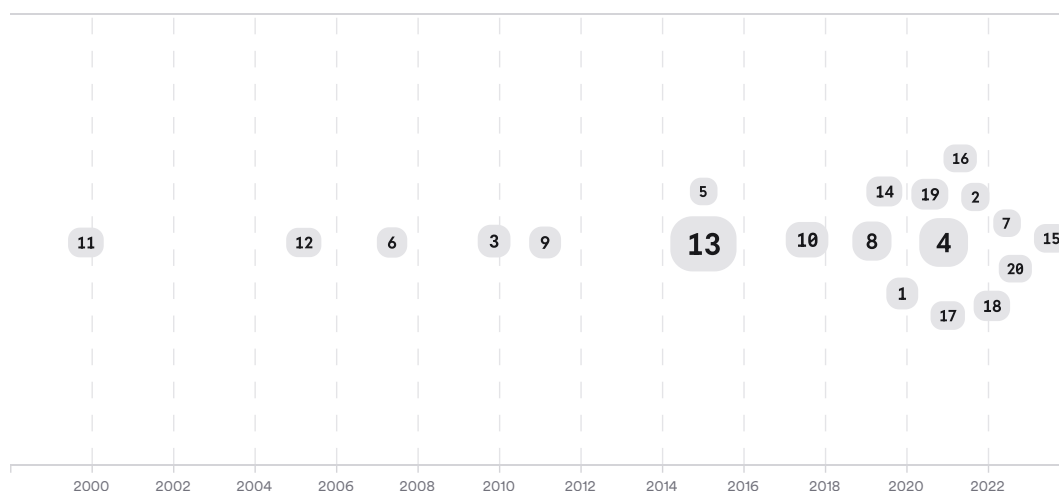


FIGURE 3 Timeline showing evolution of research on Italian pension reform strategies since the 1990s. Larger markers indicate more citations.

Top Contributors

Type	Name	Papers
Author	R. Holzmann	(Hinrichs, 2021; Castellino & Fornero, 1999; Boado-Penas et al., 2019; Franco, 2001; Murro & Palmisano, 2018)
Author	S. Gronchi	(Mazzaferro, 2021; Baldacci & Tuzi, 2003; Holzmann et al., 2019; Díaz-Saavedra et al., 2023)
Author	Felice Roberto Pizzuti	(Bridgen & Meyer, 2008; Boado-Penas et al., 2019)
Journal	<i>Social Policy & Administration</i>	(Franco & Tommasino, 2020; Gronchi et al., 2019)
Journal	<i>Journal of Pension Economics and Finance</i>	(Gronchi et al., 2023; Miles & Timmermann, 1999)
Journal	<i>Annals of Operations Research</i>	(Holzmann et al., 2019)

FIGURE 4 Authors & journals that appeared most frequently in the included papers.

4. Discussion

The shift toward an NDC PAYG system in Italy has improved long-term financial sustainability but at the cost of lower replacement rates for new generations entering the labor market after 1995 (Franco & Tommasino, 2020; Mazzaferro, 2021). To maintain living standards at retirement under these conditions requires active participation in supplementary pension schemes or increased private savings—yet uptake remains insufficient due to behavioral biases and lack of awareness among young Italians (Bongini & Cucinelli, 2019; Rinaldi, 2011). Policy efforts should focus on improving financial literacy and incentivizing voluntary contributions.

Transitioning toward a fully funded or mixed pillar system is theoretically attractive for risk diversification but faces formidable financial (transition costs/double burden) and political barriers; abrupt transitions are rare globally because they impose heavy burdens on certain cohorts (Holzmann, 2017; Lin et al., 2021; Ebbinghaus, 2015). Gradual approaches using severance pay redirection or partial payroll tax diversion have been proposed as more feasible paths but still require careful management of distributional impacts (Castellino & Fornero, 1999; Forni & Giordano, 2005).

Automatic balancing mechanisms within NDC frameworks can help address demographic shocks but may not fully guarantee adequacy without additional buffers or minimum guarantees—especially for those with fragmented careers or low earnings potential (Devolder et al., 2020). International experience suggests that while multi-pillar systems are now common across Europe as a response to aging populations and fiscal constraints (Hinrichs, 2021; Ermolieva & Shestakova, 2022), their success depends on effective regulation, communication strategies, risk-sharing mechanisms (such as minimum guarantees), and sustained political commitment.

Claims & Evidence Table

Claim	Evidence Strength	Reasoning	Papers
Supplementary pension savings are necessary for younger Italians under NDC PAYG	 Strong	Replacement rates will fall below previous DB levels; only additional savings can close the gap	(Marchionne, 2021; Capuano, 2015; Cesari et al., 2007; Pizzuti, 2022)

Claim	Evidence Strength	Reasoning	Papers
Transitioning fully from PAYG to funded pensions imposes high double-burden costs	 Strong	Current workers must finance both retirees' benefits & own assets; few countries have succeeded	(Holzmann, 2017; Andersen et al., 2021; Forni & Giordano, 2005; Ebbinghaus, 2015)
Mixed/multi-pillar systems offer better risk diversification but may lower average replacement rates	 Moderate	Diversification reduces demographic/market risks but requires higher contributions/savings	(Hinrichs, 2021; Castellino & Fornero, 1999; Boado-Penas et al., 2019)
Participation in supplementary schemes remains low due to lack of awareness/financial literacy	 Moderate	Surveys show young Italians overestimate public pensions & lack knowledge about alternatives	(Bongini & Cucinelli, 2019; Rinaldi, 2011)
Automatic balancing mechanisms can improve NDC sustainability but may not ensure adequacy alone	 Moderate	ABMs stabilize finances but don't address all adequacy/distributional issues without extra buffers	(Devolder et al., 2020)
Abrupt transitions toward funding are politically unfeasible in Italy's context	 Moderate	Political resistance arises from perceived unfairness/double burden; gradualism preferred	(Lin et al., 2021; Ebbinghaus, 2015)

FIGURE Key claims and support evidence identified in these papers.

5. Conclusion

Maintaining adequate retirement income under Italy's NDC PAYG system requires new generations to actively engage in supplementary pension saving strategies—either through occupational/individual funds or increased private savings—to offset declining replacement rates. While theoretical models support diversification via mixed pillar systems for greater sustainability and risk management, practical implementation is constrained by high transition costs ("double burden") and political realities. Policy focus should remain on improving financial literacy among youth while exploring gradual reforms that balance adequacy with fiscal responsibility.

Research Gaps

Topic/Outcome	Young Workers	Older Workers	Interrupted Careers	Supplementary Pensions	Transition Costs
Replacement rate projections	7	2	3	5	2
Supplementary scheme uptake	6	GAP	2	7	GAP
Double burden analysis	2	GAP	1	GAP	6
Welfare/distributional impact	3	1	4	2	2

FIGURE Matrix showing coverage gaps by population group/outcome versus study attribute.

Open Research Questions

Future research should explore how best to incentivize supplementary saving among youth; assess long-term outcomes under different transition scenarios; evaluate automatic balancing mechanisms' effectiveness; and examine distributional impacts across diverse worker profiles.

Question	Why
What policy interventions most effectively increase supplementary pension uptake among young Italian workers?	Understanding which incentives work best could help close projected retirement income gaps.
How do different designs for gradual transitions toward mixed pillar systems affect intergenerational equity?	Comparative analysis would inform policymakers about minimizing double-burden effects.
Can automatic balancing mechanisms ensure both sustainability and adequacy under adverse demographic scenarios?	Testing ABMs' limits is crucial given ongoing population aging risks in Italy/EU broadly.

FIGURE Open questions highlight directions for future research into Italian pension reform.

In summary: Supplementary saving is essential under Italy's reformed NDC PAYG regime; while multi-pillar diversification offers promise for long-term sustainability, practical challenges around transition costs/politics remain unresolved—requiring further research into effective incentives and equitable reform pathways.

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at <https://consensus.app>. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

References

- Andersen, T., Bhattacharya, J., & Gestsson, M. (2021). Pareto-improving transition to fully funded pensions under myopia. *Journal of Demographic Economics*, 87, 169 - 212. <https://doi.org/10.1017/dem.2020.13>
- Baldacci, E., & Tuzi, D. (2003). Demographic Trends and Pension System in Italy: An Assessment of 1990s Reforms. *Labour*, 17, 209-240. <https://doi.org/10.1111/1467-9914.17.specialissue.8>
- Boado-Penas, M., Godínez-Olivares, H., Haberman, S., & Serrano, P. (2019). Automatic balancing mechanisms for mixed pension systems under different investment strategies. *The European Journal of Finance*, 26, 277 - 294. <https://doi.org/10.1080/1351847x.2019.1647260>
- Boado-Penas, M., Eisenberg, J., & Korn, R. (2019). Transforming public pensions: A mixed scheme with a credit granted by the state. *Insurance: Mathematics and Economics*. <https://doi.org/10.1016/j.insmatheco.2020.11.005>
- Bongini, P., & Cucinelli, D. (2019). University students and retirement planning: never too early. *International Journal of Bank Marketing*. <https://doi.org/10.1108/ijbm-03-2018-0066>
- Borella, M., & Moscarola, F. (2009). Microsimulation of pension reforms: behavioural versus nonbehavioural approach. *Journal of Pension Economics and Finance*, 9, 583 - 607. <https://doi.org/10.1017/s1474747209990163>
- Bridgen, P., & Meyer, T. (2008). Politically Dominant but Socially Flawed: Projected Pension Levels for Citizens at Risk in Six European Multi-Pillar Pension Systems. 111-131. https://doi.org/10.1057/9780230227392_7
- Capuano, R. (2015). Italy's Supplementary Pension System. 81-96. https://doi.org/10.1057/9781137507594_5
- Castellino, O., & Fornero, E. (1999). From PAYG to Funding in Italy: A Feasible Transition?. *The Geneva Papers on Risk and Insurance - Issues and Practice*, 24, 473-487. <https://doi.org/10.1111/1468-0440.00035>
- Cesari, R., Grande, G., & Panetta, F. (2007). Supplementary Pension Schemes in Italy: Features, Development and Opportunities for Workers. <https://doi.org/10.2139/ssrn.1014579>

- Devolder, P., Levantesi, S., & Menziatti, M. (2020). Automatic balance mechanisms for notional defined contribution pension systems guaranteeing social adequacy and financial sustainability: an application to the Italian pension system. *Annals of Operations Research*, 299, 765-795. <https://doi.org/10.1007/s10479-020-03819-x>
- Díaz-Saavedra, J., Marimon, R., & De Sousa, J. (2023). A Worker's Backpack as an Alternative to Payg Pension Systems. *Journal of the European Economic Association*. <https://doi.org/10.1093/jeea/jvad021>
- Ebbinghaus, B. (2015). The Privatization and Marketization of Pensions in Europe: A Double Transformation Facing the Crisis. 56-73. <https://doi.org/10.18278/epa.11.5>
- Ermolieva, E., & Shestakova, E. (2022). CURRENT TRANSFORMATIONS OF PENSION SYSTEMS IN EUROPEAN AND LATIN AMERICAN COUNTRIES. *Urgent Problems of Europe*. <https://doi.org/10.31249/ape/2022.03.04>
- Forni, L., & Giordano, R. (2005). Funding a PAYG Pension System: The Case of Italy. *Fiscal Studies*, 22, 487-526. <https://doi.org/10.1111/j.1475-5890.2001.tb00050.x>
- Franco, D., & Tommasino, P. (2020). Lessons From Italy: A Good Pension System Needs an Effective Broader Social Policy Framework. *Intereconomics*, 55, 73 - 81. <https://doi.org/10.1007/s10272-020-0874-4>
- Franco, D. (2001). Italy: The Search for a Sustainable PAYG Pension System.
- Gronchi, S., Nisticò, S., & Bevilacqua, M. (2019). The Italian NDC Scheme: Evolution and Remaining Potholes. *Progress and Challenges of Nonfinancial Defined Contribution Pension Schemes: Volume 1. Addressing Marginalization, Polarization, and the Labor Market*. https://doi.org/10.1596/978-1-4648-1453-2_ch4
- Gronchi, S., Nisticò, S., & Bevilacqua, M. (2023). Increasing longevity, NDC implementation in Italy and Sweden, and all that. *European Journal of Social Security*, 25, 261 - 287. <https://doi.org/10.1177/13882627231176537>
- Hinrichs, K. (2021). Recent pension reforms in Europe: More challenges, new directions. An overview. *Social Policy & Administration*. <https://doi.org/10.1111/spol.12712>
- Holzmann, R. (2017). The ABCs of Nonfinancial Defined Contribution (NDC) Schemes. <https://doi.org/10.1111/issr.12142>
- Holzmann, R., Palmer, E., Palacios, R., & Sacchi, S. (2019). Progress and Challenges of Nonfinancial Defined Contribution Pension Schemes: Volume 1. Addressing Marginalization, Polarization, and the Labor Market. *World Bank Publications*. <https://doi.org/10.1596/978-1-4648-1453-2>
- Holzmann, R., Palmer, E., Palacios, R., & Sacchi, S. (2019). Progress and Challenges of Nonfinancial Defined Contribution Pension Schemes: Volume 2. Addressing Gender, Administration, and Communication. <https://doi.org/10.1596/978-1-4648-1455-6>
- Lin, H., Tanaka, A., & Wu, P. (2021). Shifting from pay-as-you-go to individual retirement accounts: A path to a sustainable pension system. *Journal of Macroeconomics*, 69, 103329. <https://doi.org/10.1016/j.jmacro.2021.103329>
- Marchionne, F. (2021). Una proposta di riforma per il sistema pensionistico italiano. *Moneta e Credito*, 57, 161-196.
- Mazzaferro, C. (2021). The transition to NDC in Italy: assessing distributive and financial effects. *Journal of Pension Economics and Finance*. <https://doi.org/10.1017/s1474747221000329>
- Miles, D., & Timmermann, A. (1999). Risk sharing and transition costs in the reform of pension systems in Europe. *Economic Policy*, 14, 252-286. <https://doi.org/10.1111/1468-0327.00050>
- Murro, P., & Palmisano, F. (2018). Italy: An Ageing Country with Low Level of Private Pension Schemes but High Homeownership Rate. 145-178. https://doi.org/10.1007/978-3-319-75211-2_6
- Pizzuti, F. (2022). Italy: The Italian pension system - reform trajectories and open issues. *The Evolution of Supplementary Pensions*. <https://doi.org/10.4337/9781800372986.00015>
- Poškutė, V., Gudaitis, T., Medaškis, T., & Mečkovskis, J. (2022). SEARCH FOR SUSTAINABLE PENSION SYSTEM AND STATE SUPPORT FOR FUNDED PENSIONS IN CEE COUNTRIES. *Business: Theory and Practice*. <https://doi.org/10.3846/btp.2022.16250>
- Rinaldi, A. (2011). Pension awareness and nation-wide auto-enrolment: the Italian experience.